SUPPLEMENTARY MATERIALS

Systematic review of innovative diagnostic tests for chronic exertional compartment syndrome

SUPPLEMENTARY TABLE 1 Search strategies on 31 may 2021

Search Engine	Query	Results
Pubmed	(("Compartment Syndromes"[majr:noexp] OR "Anterior Compartment Syndrome"[majr] OR "chronic exertional compartment syndrome"[ti] OR "exercise induced compartment syndrome"[ti] OR "anterior tibial syndrome"[ti] OR "medial tibial syndrome"[ti] OR "medial tibial stress syndrome"[ti] OR "chronic anterior compartment syndrome"[ti] OR "exercise related leg pain"[ti] OR "overuse lower leg injuries"[ti] OR "overuse lower leg injury"[ti] OR "exertional medial tibial pain"[ti] OR "exercise induced leg pain"[ti] OR "chronic recurrent exertional leg pain"[ti] OR "lower extremity overuse injury"[ti] OR "lower extremity overuse injuries"[ti] OR "exercise induced compartment syndrome"[ti] OR "running related musculoskeletal injuries"[ti] OR "running related musculoskeletal injury"[ti] OR "march gangrene"[ti] OR "Anterior Compartment Syndrome"[ti] OR "Medial Tibial Stress Syndrome"[majr] OR "Medial Tibial Stress Syndrome"[ti] OR "deep posterior compartment"[ti] OR "peroneal compartment"[ti]) AND ("Diagnosis"[Mesh] OR "Compartment Syndromes/diagnosis"[majr:noexp] OR "Anterior Compartment Syndrome/diagnosis"[majr] OR diagnos*[tiab]))	5,066
EMBASE	((*compartment syndrome/ OR exp *anterior tibial syndrome/ OR "Anterior Compartment Syndrome".ti. OR "chronic exertional compartment syndrome".ti. OR "exercise induced compartment syndrome".ti. OR "anterior tibial syndrome".ti. OR "medial tibial syndrome".ti. OR "medial tibial stress syndrome".ti. OR "chronic anterior compartment syndrome".ti. OR "exercise related leg pain".ti. OR "overuse lower leg injuries".ti. OR "overuse lower leg injury".ti. OR "exertional medial tibial pain".ti. OR "exercise induced leg pain".ti. OR "chronic recurrent exertional leg pain".ti. OR "lower extremity overuse injury".ti. OR "lower extremity overuse injuries".ti. OR "exercise induced compartment syndrome".ti. OR "running related musculoskeletal injuries".ti. OR "running related musculoskeletal injury".ti. OR "march gangrene".ti. OR exp *"Medial Tibial Stress Syndrome"/ OR "Medial Tibial Stress Syndrome".ti. OR "deep posterior compartment".ti. OR "peroneal compartment".ti.) AND (exp "Diagnosis"/ OR diagnos*.ti,ab OR *"compartment syndrome"/di OR exp *"anterior tibial syndrome"/di)	2,415
Web of Science	(TI=("anterior tibial syndrome" OR "Anterior Compartment Syndrome" OR "chronic exertional compartment syndrome" OR "exercise induced compartment syndrome" OR "anterior tibial syndrome" OR "medial tibial syndrome" OR "medial tibial stress	235

	syndrome" OR "chronic anterior compartment syndrome" OR "exercise related leg pain" OR "overuse lower leg injuries" OR "overuse lower leg injury" OR "exertional medial tibial pain" OR "exercise induced leg pain" OR "chronic recurrent exertional leg pain" OR "lower extremity overuse injury" OR "lower extremity overuse injuries" OR "exercise induced compartment syndrome" OR "running related musculoskeletal injuries" OR "running related musculoskeletal injury" OR "march gangrene" OR "Medial Tibial Stress Syndrome" OR "deep posterior compartment" OR "peroneal compartment") AND TS=(diagnos*))	
Cochrane	(("anterior tibial syndrome" OR "Anterior Compartment Syndrome" OR "chronic exertional compartment syndrome" OR "exercise induced compartment syndrome" OR "anterior tibial syndrome" OR "medial tibial syndrome" OR "medial tibial stress syndrome" OR "chronic anterior compartment syndrome" OR "exercise related leg pain" OR "overuse lower leg injuries" OR "overuse lower leg injury" OR "exertional medial tibial pain" OR "exercise induced leg pain" OR "chronic recurrent exertional leg pain" OR "lower extremity overuse injury" OR "lower extremity overuse injuries" OR "exercise induced compartment syndrome" OR "running related musculoskeletal injuries" OR "running related musculoskeletal injury" OR "march gangrene" OR "Medial Tibial Stress Syndrome" OR "deep posterior compartment" OR "peroneal compartment") AND (diagnos*))	15
CENTRAL	(("anterior tibial syndrome" OR "Anterior Compartment Syndrome" OR "chronic exertional compartment syndrome" OR "exercise induced compartment syndrome" OR "anterior tibial syndrome" OR "medial tibial syndrome" OR "medial tibial stress syndrome" OR "chronic anterior compartment syndrome" OR "exercise related leg pain" OR "overuse lower leg injuries" OR "overuse lower leg injury" OR "exertional medial tibial pain" OR "exercise induced leg pain" OR "chronic recurrent exertional leg pain" OR "lower extremity overuse injury" OR "lower extremity overuse injuries" OR "exercise induced compartment syndrome" OR "running related musculoskeletal injuries" OR "running related musculoskeletal injury" OR "march gangrene" OR "Medial Tibial Stress Syndrome" OR "deep posterior compartment" OR "peroneal compartment") AND (diagnos*))	15
Emcare	((*compartment syndrome/ OR exp *anterior tibial syndrome/ OR "Anterior Compartment Syndrome".ti. OR "chronic exertional compartment syndrome".ti. OR "exercise induced compartment syndrome".ti. OR "anterior tibial syndrome".ti. OR "medial tibial syndrome".ti. OR "medial tibial stress syndrome".ti. OR "chronic anterior compartment syndrome".ti. OR "exercise related leg pain".ti. OR "overuse lower leg injuries".ti. OR "overuse lower leg injury".ti. OR "exertional medial tibial pain".ti. OR "exercise induced leg pain".ti. OR "chronic recurrent exertional leg pain".ti. OR "lower extremity overuse injury".ti. OR "lower extremity overuse injuries".ti. OR "exercise induced compartment syndrome".ti. OR "running related musculoskeletal injuries".ti. OR "running related musculoskeletal injury".ti. OR "march gangrene".ti. OR exp *"Medial Tibial Stress Syndrome"/ OR "Medial Tibial Stress Syndrome".ti. OR "deep posterior compartment".ti. OR "peroneal compartment".ti.) AND (exp "Diagnosis"/ OR diagnos*.ti,ab)	905

SUPPLEMENTARY TABLE 2 Studies evaluating the use of magnetic resonance imaging (MRI) as a diagnostic test for patients with CECS in the leg.

Author	Reference	Exercise protocol	Timing MRI	Tesla Dose	MRI protocol	Main results	Conclusion
Amendola et al. [1]	ICPM (> 15 mmHg persistent elevation and failure to return to baseline values postexercise)	Treadmill	Before and 5, 15, 25 minutes after exercise	-	Each cross-sectional image was divided into the four anatomical compartments and the T1 and T2 parameters of 10 pixels were calculated.	Analyzing change in T1 and T2 values diagnosed CECS in four out of the 20 ERLP patients. All four of these patients had positive ICPM results and had abnormal MRI studies with failure of T1 to return to baseline values. One patients with a clinical history and positive ICPM results did not correlate with the MRI.	The potential usefulness of MRI as noninvasive tool in ERLP has been demonstrated.
Andreisek et al. [2]	ICPM (> 15 mmHg after exercise)	Dorsiflexion against resistance	Before and 3, 6, 9, 12 and 15 minutes after exercise	1.5	Standard, BOLD (T2* sequence) and ASL (EPI sequence) MRIs were performed in the axial plane at the level of mid-calf. T2* and ASL signal changes in ROI over time were calculated relative to the baseline examination.	In both patients and control subjects, changes in T2* and ASL showed a peak at 3 minutes after exercise, followed by a decrease over time. No statistically significant differences between patients and control subjects in postexercise changes were found.	After calf muscle exercise, no statistically significant differences in T2* relaxation times or ASL signal, indicative of differences in muscle oxygenation and perfusion status, were found between patients with chronic exertional

							compartment syndrome and control subjects.
Eskelin et al. [3]	ICPM (> 40 mmHg 1 min after exercise)	Treadmill	Before and after exercise	0.1	The imaging sequence was a T2-weighted spin-echo sequence. The transaxial sections through the lower leg were visually surveyed, and five ROIs were outlined for signal intensity measurement. The pre- and postexercise sections were drawn side by side on the screen to make the ROI areas as similar as possible.	In the patient group, the relative change from rest to the postexercise state in the normalized MR signal intensity parameter correlated significantly with the respective change in intracompartmental pressure and with the absolute postexercise pressure. In the patients with elevated postexercise intracompartmental pressure, the increase in normalized MRI signal intensity from rest to the postexercise state was significantly greater than that in the control subjects or the patients with normal or borderline postexercise intracompartmental pressure.	MRI performed at rest and immediately after muscular exercise is a promising method for diagnosing chronic exertional compartment syndrome and assessing its severity.
Kiuru et al. [4]	ICPM (> 40 mmHg 1	n/a	At rest	1.0	Routine T1-weighted spin echo, T2-	There was no statistically significant	In cases where CECS is suspected clinically,

	min after exercise)				weighted fast spin echo with fat suppression, and STIR sequences were obtained. The average signal intensity of three ROIs was calculated.	difference in findings relating to bone between patients with and without CECS. No statistically significant difference in signal intensity ratios relating to anterior and superficial posterior compartmental muscles between patients with and without CECS was found.	ICPM should be measured and MRI undertaken to reveal any bone abnormalities.
Litwiller et al. [5]	ICPM (n/a) or clinical outcome	Dorsi- or plantar flexion against resistance	During isometric contraction	1.5	The imaging protocol consisted of T1-weighted images, T2-weighted sequences and, if indicated, a fast spin echo imaging with chemical fat suppression. Polygonal ROIs were analyzed.	Receiver operating characteristic analysis showed a threshold for the ratio of relative T2-weighted signal intensity of 1.54 to have a sensitivity of 96%, specificity of 90% and accuracy of 96% for CECS. Patients with CECS had their peak ratio of signal intensity compared with baseline during the first recovery period after isometric dorsi flexion, whereas unaffected subjects and ERLP patients with	With the presented MRI protocol high accuracy, sensitivity and specificity for CECS diagnosis was shown, making it an eligible triage tool for ICPM.

						reached their peak values during exercise.	
Ringler et al. [6]	ICPM (Pedowitz)	Dorsi- or plantar flexion against resistance	During isometric contraction	1.5	The imaging protocol consisted of an initial acquisition of T1-weighted images, followed by separate T2-weighted acquisitions. Polygonal ROIs were analyzed. The previously established optimal T2-weighted signal intensity ratio of 1.54 was considered the diagnostic threshold for the validation.	Sensitivity and specificity were 96% (95% CI: 79–99%) and 87% (95% CI: 75–94%) using the established threshold of 1.54. T2-weighted intensity ratio provided excellent discrimination with a concordance statistic of 0.96 (95% CI: 0.91–1.00).	In-scanner exercise-based MRI demonstrated reliability and reproducibility as a noninvasive screening test for CECS, thus reducing the need for invasive ICPM.
Sigmund et al. [7]	n/a	Treadmill	At rest and 10 minutes after exercise	-	Spectral adiabatic inversion recovery T2w and diffusion tensor imaging were obtained. Increased T2w signal and diffusion tensor imaging values on post-exercise images were evaluated as by the use of ROIs.	All diffusivities significantly increased and fractional anisotropy decreased with exercise. Longitudinal diffusion responses were significantly smaller than transversal diffusion responses. Nineteen of 98 patient compartments were classified as CECS on T2w.	Diffusion tensor imaging shows promise as an ancillary imaging method in the diagnosis of CECS.

Van den Brand et al. [8]	ICPM (≥35mmHg 1 min after exercise)	Treadmill	At rest and 1 minute after exercise	-	In each ROI, the average T2-weighted signal intensity was calculated, after which the average value of the 3 ROIs was arbitrarily accepted as the average T2-weighted signal intensity in the entire cross section of the compartment.	The percentage increase in average T2-weighted signal intensity ratio during exercise (anterior compartment–superficial posterior compartment) was higher, albeit not statistically significant, in those with CECS, as was the percentage increase in the anterior compartment.	MRI is clearly less suitable to confirm CECS, compared to NIRS and ICPM.
Verleisdonk et al. [9]	ICPM (≥ 35mmHg 1 min after exercise)	Treadmill	At rest and 1 minute after exercise	1.5	T2-weighted spin echo sequences were obtained. Three ROIs were chosen in the cross-sections in both the anterior and superficial posterior compartments. At least 3 months after fasciotomy, on the basis of the same protocol, MRI and tissue pressure measurement were repeated.	The signal intensity ratio between the anterior and posterior compartments increased by 21.0% (range 12.2–32%) after exercise in patients with a chronic compartment syndrome. After fasciotomy, this ratio increased after exercise by a mere 1.8% (range 0–8%). The difference between the increases before and after fasciotomy is statistically significant. The difference in the increases in controls and	In view of the substantial increase in T2-weighted signal intensity, MRI can be used in diagnosing chronic compartment syndrome.

						patients with a compartment syndrome before fasciotomy was statistically significant.	

ASL = arterial spin labeling, CECS = chronic exertional compartment syndrome, EPI = echo-planar imaging, ERLP = exercise-related leg pain, ICPM = intracompartmental pressure measurement, ROI = region of interest, STIR = short tau inversion recovery.

SUPPLEMENTARY TABLE 3 Studies evaluating the use of near infrared spectrometry (NIRS) as a diagnostic test for patients with CECS in the leg.

Author	Reference	Exercise protocol	Type of probe	Probe position	NIRS protocol	Main results	Conclusion
Mohler et al. [10]	ICPM (Pedowitz)	Dorsi- or plantar flexion against resistance	RunMan, NIM	On the surface of the leg over the middle third of the anterior compartment.	Changes in oxygenation were measured continuously during an isokinetic exercise protocol.	CECS patients had greater relative deoxygenation than did the ERLP patients and the control subjects. Similarly, maximum relative deoxygenation was greater in CECS patients compared to ERLP patients and control subjects. CECS patients had greater relative deoxygenation during the early part of the recovery period and a longer interval until recovery of the pre-exercise resting level than did the other two	The results of the present study suggest that near-infrared spectroscopy may be useful as a noninvasive diagnostic tool for the evaluation of patients suspected of having chronic anterior compartment syndrome of the leg.

						groups.	
Rennerfelt et al. [11]	ICPM (Pedowitz)	Treadmill or dorsiflexion against resistance	InSpectra Tissue Spectrometer	Central over the anterior tibial muscle.	StO_2 values were collected continuously every 3.5 seconds throughout the measurements. Each measurement was marked to indicate the time for the start of exercise, the end of exercise, and the end of recovery.	The sensitivity was 34% and the specificity was 43% when an StO_2 level of ≤8% at peak exercise was used to indicate CECS. The sensitivity and the specificity were only 1% when an StO_2 level of ≤ 50% at peak exercise was used to indicate CECS. When thirty seconds or more at T90 was set as the cut-off value for a prolonged time for reoxygenation, indicating a diagnosis of CECS, the sensitivity was 38% and the specificity was 50%.	Changes in muscle oxygen saturation during and after an exercise test that elicits leg pain cannot be used to distinguish between patients with CECS or other ERLP patients.
Van den Brand et al. [8]	ICPM (>35mmHg 1 minute after exercise)	Treadmill	InSpectra Tissue Spectrometer	Directly on the skin over the middle one third of the anterior compartment.	During and after exercise, StO2 in the affected compartment was continuously registered. The reflected recollected near-infrared light	The ICP and NIRS values were significantly different in legs in which CECS had been diagnosed in comparison to those without. The best estimate of sensitivity	The sensitivity of noninvasive near-infrared spectroscopy is clinically equivalent to that of invasive intracompartmental pressure measurements.

					can be used to provide the StO_2 value of the underlying part of the body.	for NIRS found in this study was 85% (76%–92%, exact 95% CI), which validates the previous estimate of 90%.	
Van den Brand et al. [12]	ICPM (> 35mmHg 1 minute after exercise)	Treadmill	InSpectra Tissue Spectrometer	On the skin over the tibialis anterior muscle, approximately 10 cm distal to the tibial tuberosity and 2 cm lateral of the tibial crest.	StO_2 measurements were collected continuously every 3.5 seconds throughout the entire protocol.	Most of the StO_2 end points also showed a highly significant difference between the healthy volunteers and those with CECS and between the preoperative and postoperative values. The slower recovery of tissue saturation in CECS patients in comparison to healthy volunteers as reported in earlier publications was not found. When tested for their sensitivity and specificity, different cut-off values for various StO2 parameters or combinations of parameters proved to be very comparable to	StO_2 measured through NIRS is a noninvasive, painless, easy-to-use alternative to ICP in the diagnosis of CECS.

						those found for ICP	
Zhang et al. [13]	ICPM (>30 mmHg after exercise)	Dorsi- or plantar flexion against resistance	RunMan, NIM	centrally over the tibialis anterior muscle of the most symptomatic leg.	The relative change of oxygenation in the tibialis anterior muscle was measured continuously noninvasively before, during, and after each exercise test. Levels of intramuscular oxygenation during exercise and reoxygenation after exercise were monitored.	The mean level of oxygenation decreased to 33±19% in patients with CACS and to 34±19% in patients without CACS compared with baseline level (100%) at rest before exercise. The level of oxygenation was below 20% during the exercise test in 12 of 47 (26%) patients with CACS and in 30 of 129 (23%) patients without CACS. The sensitivity was 60% and specificity was 45% if R90 ≥30 seconds was selected as the cut-off value indicating a diagnosis of CECS.	We conclude that the magnitude of intramuscular deoxygenation during exercise is an unreliable measure to diagnose CECS. However, the time for reoxygenation returning to baseline level following an exercise test is a valuable adjunct in diagnosing CECS.

CECS = chronic exertional compartment syndrome, CACS = chronic anterior compartment syndrome, ERLP = exercise-related leg pain, ICPM = intracompartmental pressure measurement,

SUPPLEMENTARY TABLE 4 Studies with single-photon emission computerized tomography (SPECT)

Author	Reference (cut-off)	Exercise protocol	Timing SPECT	Injection material	Other SPECT specifics	Main results	Conclusion
Allen et al. [14]	ICPM (exercise pressure > 40mmHg)	n/a	n/a	400 MBq of ^{99m}Technetium-Medronate	Three-phase bone scans were obtained immediately, 3 minutes and 3 hours post-injection.	A distinctive pattern of uptake was seen in 30 patients, 24 of whom had MTSS and 6 of whom had CECS. Of the patients with normal scans, only 4 had MTSS, the remaining 15 had CECS. Both of these findings are statistically significant.	Bone scans are a useful diagnostic tool in the differential diagnosis of ERLP.
Edwards et al. [15]	ICPM (n/a) or clinical outcome	Treadmill	At rest and subsequent to peak exercise	300 MBq of ^{99m}Technetium-MIBI	Tomographic imaging was performed immediately using a 360° rotation for both stress and rest imaging. The imaging criterion was a visually detectable decrease in Technetium uptake in one or more compartments of the exercise study when compared with the rest study.	Statistical analysis gave a p=0.06 when comparing ^{99}Tc^{m}-MIBI to ICPM results. Comparing ^{99}Tc^{m}-MIBI to clinical outcome gave a $P < 0.0001$ with a sensitivity of 80% and specificity 97%. The positive predictive value was 89% and the negative predictive value 94%.	^{99}Tc^{m}-MIBI can detect compartment syndromes with good positive and negative predictive values. This technique shows promise in the diagnosis of CECS.
Hayes et al. [16]	ICPM	Provocative	At rest and	100 MBq of	Imaging was	In this study, Thallium	These preliminary

	(n/a)	exercises	subsequent to peak exercise	^{201}Thallium-chloride	commenced 5 min postinjection and repeated after a 4-hr delay. Percent change in activity between exercise and rest was determined by defining each leg separately on the curves and calculating the area under the curve. Redistribution was defined as a qualitative or quantitative increase in uptake or counts relative to the stress image. Washout was defined as a qualitative or quantitative reduction in uptake or counts compared to the stress image.	demonstrated reversible ischemia in the four patients with typical symptoms of CPS and elevated compartment pressure measurements. In contrast, the control subjects showed uniform uptake in the early images and washout in the delayed images. Additionally, a patient showed definite improvement postoperatively making a diagnosis of CPS with only borderline elevated compartment pressure most likely.	data are encouraging and would indicate the potential use of Thallium in the investigation of CPS. Thallium-201 can demonstrate the pathophysiology of CPS, i.e., that elevated pressure reduces local blood flow in the calves resulting in ischemia.
Oturai et al. [17]	ICPM (Pedowitz)	Treadmill	Before and after exercise	367 MBq of ^{99m}Technetium-Tetrofosmin	Both rest and stress images were acquired with a	Six subjects had increased pressures indicating the	Tc-99m-tetrofosmin perfusion SPECT seems not to be a

					rectangular field of view dual-headed camera system using low-energy high-resolution parallel-hole collimators. ROIs were used to demarcate relevant compartments. Scoring each ROI for changes in relative activity between rest and stress suggesting stress-induced relative hypoperfusion was performed.	presence of CECS. In three (50%) of these muscular hypoperfusion was observed by perfusion SPECT. In eight subjects with normal ICP's SPECT suggested muscular hypoperfusion. This gives a sensitivity of 50% and specificity of 63%.	useful method for diagnosing CECS.
Takebayashi et al. [18]	ICPM (Pedowitz)	Treadmill	After exercise	74 MBq of ^{201}Thallium-Choloride	SPECT images were obtained pre- and postoperatively, at 5 and 180 minutes post injection as stress and delayed images respectively. In each compartment, the mean	The SPECT images demonstrated 16 ischemic compartments in eight of the nine patients. The SPECT results were consistent with those of quantitative analysis. There were discrepancies between the clinical and SPECT	The present study suggests that ^{210}Thallium-chloride SPECT has promise for diagnosing and localizing CECS.

					percentage of the maximum uptake was calculated after ROIs were drawn manually over each compartment.	diagnoses in six legs (33% of the 18 legs) of five patients. Postoperative SPECT demonstrated ^{201}TIperfusion was improved in all involved compartments for that fasciotomy was performed.	
Trease et al. [19]	ICPM (Pedowitz)	Compartment-specific exercises	After exercise	0.8 MBq of ^{201}Thallium-Choloride per kg body weight	SPECT images were obtained 5 and 180 minutes postinjection as stress and delayed images respectively. ROI were used to calculate the percentage of pixel count.	There was no significant difference in either quantitative or qualitative assessments of perfusion between those compartments with and those without CECS.	This study was unable to reproduce the results of early smaller studies using ^{201}Tl SPECT imaging, which had suggested a diagnostic role for ^{201}Tl SPECT imaging in CECS.

CECS = chronic exertional compartment syndrome, ERLP = exercise-related leg pain, ICPM = intracompartmental pressure measurement, MIBI = methoxyisobutyl isonitrile, ROI = region of interest.

SUPPLEMENTARY TABLE 5 Studies evaluating the use of diagnostic tests other than MRI, NIRS or SPECT for patients with CECS.

Author	Reference	Diagnostic test	Protocol	Main results	Conclusion
Abraham et al. [20]	ICPM (>40 mmHg 1	Laser Doppler flow (LDF)	A sterilised flexible optic fibre 10 to 15 cm long	The LDF after exercise tended to be higher in	There is a different time course in the LDF findings in

	minute after exercise)		with LDF-meter was introduced into the muscle, close to the site of the ICPM, for continues recording of microcirculatory muscle blood flow after exercise.	controls than in CECS patients after one minute. During the remainder of the recovery period the LDF remained lower in control subjects than in patients with CCS. The difference in LDF between the two groups was statistically significant from 4 to 16 minutes, but not after this.	control subjects and patients with CECS. LDF can be used to help in the diagnosis.
Burnham et al. [21]	ICPM (> 35 mmHg 1 minute after exercise)	Nerve conduction	Using surface electrodes, nerve conduction of the bilateral deep peroneal motor, superficial peroneal sensory, and tibial motor nerves were performed immediately after the ICPM. Distal latency, conduction velocity, wave form amplitude (negative peak), and conduction block were recorded.	No exercise-induced impairment of function was found of the nerves within compartments with high pressures as compared to those of normal pressure. No correlation was found between postexercise changes in electrophysiologic function and compartment pressure.	Nerve conduction studies are not useful in identifying CECS of the lower leg.
de Bruijn et al. [22]	ICPM (Pedowitz)	Nomogram	Associations of patient-related variables and the diagnosis CECS were determined using univariable logistic regression. Only variables that were present in >	Age, gender, bilateral symptoms, previous lower leg pathology, sports (running and skating) and tender muscle compartments were identified as independent predictors of	The predictive model, displayed as a nomogram, may aid in selecting individuals requiring an invasive dynamic ICPM.

			50% of the patients were included. Significant variables were included as covariates in a multivariable logistic regression model.	lower leg CECS. The proposed predictive model has moderate discriminative ability and good calibration over the complete range of predicted probabilities.	
Fouasson-chailloux et al. [23]	ICPM (Pedowitz)	Predictive clinical parameters	Absence of pain at rest and on palpation, bilateral pain, muscle hardness, stopping exercise due to pain, pain recidivism for the same exercise, pain irradiation, paraesthesia, muscle hernia at rest and after exercise, post-effort muscle weakness and muscle hypertrophy were systematically collected on a clinical yes/no questionnaire before and after the treadmill test.	Muscle hardness and weakness, muscle hernia at rest and after exercise, were predictive of CECS. Pain at rest and during palpation, and absence of pain recidivism for the same exercise were predictive of an absence of CECS.	This study confirmed the importance of clinical parameters to diagnose CECS. A better knowledge of the relevant parameters could help physicians to indicate invasive examinations.
Korhonen et al. [24]	ICPM (n/a)	EMG and myotonometry	Simultaneously with IMP measurements, EMG of the muscle was recorded using bipolar Ag/AgCl electrodes, attached to the skin surface above the muscle of interest and the mechanical characteristics of the muscle were determined	The EMG amplitude and Myotonometry frequency accounted significantly for the variations in the IMP under short-term isometric loading. All increased linearly with the relative muscle load. Mean values of EMG amplitudes at the contraction levels of 75%	Both the surface EMG and myotonometry are indicative of IMP and could be used as guiding tools in the diagnosis of CECS. However, due to the significant patient-to-patient variations, we believe that neither of these methods can be used alone.

			using myotonometry. Measurements were conducted at rest and during short-term and long-term isometric contractions.	and 100% maximum voluntary contraction and myotonometry frequency values at all contraction levels were higher for subjects with pathological values of IMP than for those with IMP values in the normal range. Total changes in IMP and EMG amplitude during 1 min isometric contraction were linearly interrelated.	
Rennerfelt et al. [25]	ICPM (Pedowitz)	Patient pain drawing	Patients received a questionnaire including an outline of a human body with anterior, posterior and lateral views. The patients indicated the afflicted areas of pain on the drawing and either single locations or larger regions could be marked. Two independent orthopaedic surgeons diagnosed the causes of leg pain based only on the PPD at least 1 year after admission.	The sensitivity of the PPD to identify CECS correctly was 67% (observer 1) and 75% (observer 2). The specificity was 65% and 54%, respectively. The interobserver agreement (n = 477) was 80%, and the kappa value was 0.55.	PPD might be a valuable instrument in diagnosing the causes of exercise-induced leg pain.
Rowdon et al. [26]	ICPM	Nerve	The recording surface	After exercise there was no	Athletes with this syndrome

	(n/a)	conduction	electrode was placed over the extensor digitorum brevis muscle belly and the reference electrode placed distal to that, over the fourth toe extensor tendon. The ground was placed between the recording electrode and the points of stimulation. Response amplitudes and latencies were recorded, and conduction velocities were calculated. Testing was performed preexercise and then immediately after exercise for the controls, and immediately after ICPM for the subjects.	difference in the nerve conduction velocities between groups, nor was there a change in conduction velocity within groups as compared with the preexercise condition. CECS patients demonstrate a mild impairment in vibratory sensation in the deep peroneal distribution, and a decreased postexercise potentiation of the peroneal motor amplitude.	may be reassured that damage to the deep peroneal nerve is minimal, although there appears to be a mild impairment in large fiber sensory function, which may contribute to the athlete's sense of loss of foot control.
Willey et al. [27]	n/a	Korotkoff sounds	Korotkoff sounds were detected over the anterior tibial artery before and after exercise.	Korotkoff sounds were audible after exercise in all 7 CECS patients and persisted longer. A slight and transient rise in pressure was observed in only 4 of 15 ERLP patients.	The method described is simple but valuable in the diagnosis of CECS.
Zhang et al. [28]	ICPM (> 30 mmHg after exercise)	EMG	A simultaneous EMG recording was obtained by using bipolar surface	In 13 of the 16 CECS patients, the EMG signal was silent, confirming that	In conclusion, our study shows that simultaneous measurements of IMP and

			electrodes placed over the anterior tibial muscle about 15cm below the knee joint. A reference electrode was placed over the lateral malleolus. The EMG signal was recorded during a maximal voluntary contraction before the exercise test, entailing maximal voluntary electromyographic activity, which was picked up by the surface electrodes. The EMG signal was also recorded continuously during and after the exercise test.	they had CCS. In the remaining three patients, the EMG signal was initially positive (5–34% of maximal voluntary contraction), and then became silent after communicating with the patients to relax their legs	EMG differentiate elevated IMP at rest after exercise due to the volume load of the compartment muscles seen in patients with CCS and due to remaining muscle contraction. There is a risk of making a false diagnosis of CCS if elevated IMP at rest after exercise is used as the only diagnostic criterion.

CECS = chronic exertional compartment syndrome, EMG = electromyography, ERLP = exercise-related leg pain, ICPM = intracompartmental pressure measurement, LDF = laser doppler flow

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